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Thermodynamics and Statistical Physics

Spring 2026

Homework 9

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Show all essential steps in your derivations.

Raoult's law for an ideal solution ★★

Consider an ideal binary solution of species A and B , in the sense defined in the preceding problem. The particle numbers are N_A and N_B , the total particle number is

$$N = N_A + N_B,$$

and the mole fractions are

$$x_A = \frac{N_A}{N}, \quad x_B = \frac{N_B}{N}.$$

According to the preceding problem, the Helmholtz free energy of the solution is

$$F(T, V, N_A, N_B) = F_{\text{pure}}(T, V, N) + T \left[N_A \log x_A + N_B \log x_B \right],$$

where $F_{\text{pure}}(T, V, N)$ is the Helmholtz free energy of the corresponding one-component pure system with N particles.

1. Carry out the Legendre transform with respect to V , and obtain the Gibbs free energy of the solution. Show that

$$G(T, P, N_A, N_B) = G_{\text{pure}}(T, P, N) + T \left[N_A \log x_A + N_B \log x_B \right],$$

where $G_{\text{pure}}(T, P, N)$ is the Gibbs free energy of the corresponding one-component pure system.

2. Show that the chemical potential of species A in the solution is

$$\begin{aligned} \mu_A(T, P, x_A) &= \left(\frac{\partial G}{\partial N_A} \right)_{T, P, N_B} \\ &= \mu(T, P) + T \log x_A, \end{aligned} \tag{1}$$

where

$$\mu(T, P) = \left(\frac{\partial G_{\text{pure}}}{\partial N} \right)_{T, P}$$

is the chemical potential of the corresponding pure substance at the same T and P .

In particular, setting $x_A = 1$ gives

$$\mu_A(T, P, x_A = 1) = \mu(T, P),$$

as it should.

Similarly, show that

$$\mu_B(T, P, x_B) = \mu(T, P) + T \log x_B.$$

3. Now consider the dilute limit

$$x_B = x \ll 1,$$

where the solution may be regarded as a dilute solution with B as the solute and A as the solvent. Show that Eq. (1) reduces to

$$\mu_A(T, P, x_A) = \mu(T, P) - Tx + O(x^2).$$

Compare this result with what you learn in the lectures.

4. Now suppose that the solution is in equilibrium with an ideal gas mixture containing vapor of A . Let the partial pressure of A in the vapor be P_A , and let the total pressure of the gas mixture be P . The chemical potential of A in the ideal gas mixture is

$$\mu_A^{\text{vap}}(T, P_A) = T \log \frac{P_A \lambda_T^3}{T}.$$

Equivalently,

$$\mu_A^{\text{vap}}(T, P_A) = \mu^{\text{IG}}(T, P) + T \log \frac{P_A}{P},$$

where

$$\mu^{\text{IG}}(T, P) = T \log \frac{P \lambda_T^3}{T}$$

is the chemical potential of a one-component ideal gas at temperature T and pressure P .

Let P_A^* be the vapor pressure of pure liquid A at the same temperature. Thus P_A^* is determined by the equilibrium condition

$$\mu(T, P) = \mu_A^{\text{vap}}(T, P_A^*).$$

For the solution, phase equilibrium requires

$$\mu_A(T, P, x_A) = \mu_A^{\text{vap}}(T, P_A).$$

Using the result of part (b), show that

$$P_A = x_A P_A^* = (1 - x_B) P_A^*.$$

This is Raoult's law for the vapor pressure of the solvent in an ideal solution. It says that the vapor pressure of A decreases linearly with its mole fraction in the liquid solution.

Solution

1. We start with the Helmholtz free energy for the ideal binary solution:

$$F(T, V, N_A, N_B) = F_{\text{pure}}(T, V, N) + T \left[N_A \log x_A + N_B \log x_B \right]$$

To obtain the Gibbs free energy G , we perform a Legendre transform with respect

to V :

$$G = F + PV$$

Substituting the expression for F , we get:

$$G(T, P, N_A, N_B) = F_{\text{pure}}(T, V, N) + PV + T \left[N_A \log x_A + N_B \log x_B \right]$$

For the corresponding pure system of N particles, the Gibbs free energy is strictly given by $G_{\text{pure}}(T, P, N) = F_{\text{pure}}(T, V, N) + PV$. Replacing this in the equation above directly yields the Gibbs free energy of the solution:

$$G(T, P, N_A, N_B) = G_{\text{pure}}(T, P, N) + T \left[N_A \log x_A + N_B \log x_B \right]$$

2. To find the chemical potential of species A , we take the partial derivative of G with respect to N_A , keeping T, P, N_B constant. Note that $N = N_A + N_B$, so $\partial N / \partial N_A = 1$.

$$\begin{aligned} \mu_A(T, P, x_A) &= \left(\frac{\partial G}{\partial N_A} \right)_{T, P, N_B} \\ &= \left(\frac{\partial G_{\text{pure}}}{\partial N} \right)_{T, P} \frac{\partial N}{\partial N_A} + T \frac{\partial}{\partial N_A} \left[N_A \log \left(\frac{N_A}{N} \right) + N_B \log \left(\frac{N_B}{N} \right) \right] \end{aligned}$$

The first term is simply $\mu(T, P)$. We expand the second term:

$$\begin{aligned} \frac{\partial}{\partial N_A} \left[N_A \log N_A - N_A \log N + N_B \log N_B - N_B \log N \right] &= \\ \left(\log N_A + 1 \right) - \left(\log N + N_A \frac{1}{N} \right) + 0 - \left(N_B \frac{1}{N} \right) &= \\ \log N_A + 1 - \log N - \frac{N_A + N_B}{N} &= \log \left(\frac{N_A}{N} \right) + 1 - 1 = \log x_A \end{aligned}$$

Combining both parts, we obtain:

$$\mu_A(T, P, x_A) = \mu(T, P) + T \log x_A$$

By perfectly symmetric mathematical steps (differentiating with respect to N_B), we get:

$$\mu_B(T, P, x_B) = \mu(T, P) + T \log x_B$$

3. Consider the dilute limit where $x_B = x \ll 1$. Since $x_A + x_B = 1$, we have $x_A = 1 - x$. Substituting this into the expression for μ_A :

$$\mu_A(T, P, x_A) = \mu(T, P) + T \log(1 - x)$$

Using the Taylor expansion $\log(1 - x) = -x - \frac{x^2}{2} - \dots$ for small x , we find:

$$\mu_A(T, P, x_A) = \mu(T, P) - Tx + O(x^2)$$

Comparison: This correctly reproduces the colligative property formula for the lowering of the solvent chemical potential in a dilute ideal solution. The addition of a small amount of solute B strictly lowers the chemical potential of the solvent A by an amount linearly proportional to the solute mole fraction x and the temperature T .

4. In phase equilibrium between the pure liquid A and its vapor, the chemical potentials must be equal:

$$\mu(T, P) = \mu_A^{\text{vap}}(T, P_A^*) = T \log \frac{P_A^* \lambda_T^3}{T}$$

For the solution in equilibrium with the gas mixture, phase equilibrium for species A requires $\mu_A(T, P, x_A) = \mu_A^{\text{vap}}(T, P_A)$. Substituting the derived forms:

$$\mu(T, P) + T \log x_A = T \log \frac{P_A \lambda_T^3}{T}$$

Substitute the pure equilibrium expression $\mu(T, P) = T \log \frac{P_A^* \lambda_T^3}{T}$ into the equation above:

$$T \log \frac{P_A^* \lambda_T^3}{T} + T \log x_A = T \log \frac{P_A \lambda_T^3}{T}$$

Dividing by T and combining the logarithms on the left-hand side:

$$\log \left(x_A \frac{P_A^* \lambda_T^3}{T} \right) = \log \frac{P_A \lambda_T^3}{T}$$

Exponentiating both sides yields Raoult's law directly:

$$P_A = x_A P_A^* = (1 - x_B) P_A^*$$

Coexistence in a binary mixture: closed system and open beaker

The phase diagram of a solution of A in B, at a pressure of 1 atm, is shown in Fig. 5.3 of the lecture notes. The horizontal axis is the mole fraction of A, denoted by x_A . The upper bounding curve of the two-phase region is

$$T = T_0 - (T_0 - T_1)x_A^2,$$

and the lower bounding curve is

$$T = T_0 - (T_0 - T_1)x_A(2 - x_A).$$

The lower curve is the bubble-point curve, giving the composition of the liquid phase at the onset of boiling. The upper curve is the dew-point curve, giving the composition of the vapor phase. Thus, at a fixed temperature inside the two-phase region, a horizontal tie line connects the liquid composition x_A^ℓ to the vapor composition x_A^v .

Except where otherwise stated, assume that the initial liquid composition is

$$x_A = \frac{1}{2}.$$

Closed system. First consider a closed system at fixed pressure 1 atm. The vapor remains in the system, and the volume is allowed to change. The total mole fraction of A is therefore conserved.

1. What is the boiling temperature?
2. What is the composition of the vapor as it first begins to boil off? Does boiling tend to increase or decrease the mole fraction of A in the remaining liquid?
3. At the moment when half of the particles are in the vapor phase, what are the mole fractions of A in the liquid phase and in the vapor phase?
4. What is the mole fraction of A in the liquid phase right before the boiling is finished?

Open beaker. Now consider an open beaker. The vapor produced by boiling escapes immediately, so the total number of particles remaining in the beaker changes during the process.

5. Let N be the number of particles remaining in the liquid phase, and let x_A be the mole fraction of A in this remaining liquid. Suppose a small amount $-dN > 0$ of liquid is boiled off. Show that

$$dx_A = - \left[(2x_A - x_A^2)^{1/2} - x_A \right] \left(\frac{-dN}{N} \right).$$

6. Solve the differential equation in part (e), with the initial condition

$$x_A = \frac{1}{2}, \quad N = N_0.$$

Express the result as an implicit relation between x_A and N/N_0 .

Hint. Use the substitution

$$x_A = 1 - \cos \theta, \quad \sqrt{x_A(2 - x_A)} = \sin \theta.$$

It is also useful to set

$$t = \tan \frac{\theta}{2}.$$

7. In the open beaker, at the moment when half of the initial liquid has boiled off, what are the mole fractions of A in the remaining liquid and in the vapor being produced at that moment? Compare with the closed-system answer.
8. According to the differential equation, does x_A go to zero only when $N \rightarrow 0$, or does it reach zero before boiling is finished?
9. Explain why the differential equation in part (e) can no longer be used after $x_A = 0$. What kind of theory is needed to describe the remaining boiling process?

Solution

Closed system. Let the composition bounds be parameterized by $\Delta T = T_0 - T_1$. The dew-point and bubble-point curves are $T_{\text{dew}} = T_0 - \Delta T x_A^2$ and $T_{\text{bubble}} = T_0 - \Delta T x_A(2 - x_A)$. The overall composition is $x_A = 1/2$.

1. **Boiling temperature:** The liquid begins to boil when the temperature is raised to the bubble-point curve evaluated at the overall initial composition $x_A^\ell = 1/2$:

$$T_{\text{boil}} = T_{\text{bubble}}(1/2) = T_0 - (T_0 - T_1) \frac{1}{2} \left(2 - \frac{1}{2} \right) = T_0 - \frac{3}{4}(T_0 - T_1) = \frac{1}{4}T_0 + \frac{3}{4}T_1$$

2. **Vapor composition at first boil:** The first bubble of vapor formed at T_{boil} must lie on the dew-point curve at the same temperature. Thus, we solve $T_{\text{dew}}(x_A^v) = T_{\text{boil}}$:

$$T_0 - (T_0 - T_1)(x_A^v)^2 = T_0 - \frac{3}{4}(T_0 - T_1) \implies (x_A^v)^2 = \frac{3}{4} \implies x_A^v = \frac{\sqrt{3}}{2} \approx 0.866$$

Because $x_A^v > x_A^\ell = 0.5$, the vapor is much richer in A than the liquid. Therefore, boiling preferentially removes A, **decreasing** the mole fraction of A in the remaining liquid.

3. **Half of the particles in vapor phase:** By the lever rule, if $N^\ell = N^v = N^{\text{tot}}/2$, the overall composition is exactly the average of the two phases: $x_A^\ell + x_A^v = 2(1/2) = 1$, which means $x_A^v = 1 - x_A^\ell$. A tie line connects them, so $T_{\text{bubble}}(x_A^\ell) = T_{\text{dew}}(x_A^v)$:

$$T_0 - \Delta T x_A^\ell(2 - x_A^\ell) = T_0 - \Delta T(1 - x_A^\ell)^2 \implies x_A^\ell(2 - x_A^\ell) = (1 - x_A^\ell)^2$$

Expanding yields $2x_A^\ell - (x_A^\ell)^2 = 1 - 2x_A^\ell + (x_A^\ell)^2 \implies 2(x_A^\ell)^2 - 4x_A^\ell + 1 = 0$. Solving this quadratic yields $x_A^\ell = \frac{4 \pm \sqrt{16-8}}{4} = 1 \pm \frac{\sqrt{2}}{2}$. Since $x_A^\ell \leq 1$, we take the minus

sign:

$$x_A^\ell = 1 - \frac{\sqrt{2}}{2} \approx 0.293, \quad x_A^v = \frac{\sqrt{2}}{2} \approx 0.707$$

4. **Last drop of liquid:** Right before boiling finishes, almost all the system is vapor, so $x_A^v \approx 1/2$. The remaining drop of liquid lies on the bubble curve connected by a tie line: $T_{\text{bubble}}(x_A^\ell) = T_{\text{dew}}(1/2)$.

$$x_A^\ell(2 - x_A^\ell) = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \implies (x_A^\ell)^2 - 2x_A^\ell + \frac{1}{4} = 0$$

Solving this gives $x_A^\ell = \frac{2 \pm \sqrt{4-1}}{2} = 1 \pm \frac{\sqrt{3}}{2}$. Since $x_A^\ell \leq 1$:

$$x_A^\ell = 1 - \frac{\sqrt{3}}{2} \approx 0.134$$

Open beaker.

5. **Differential Equation:** Let N be the liquid particle number and x_A its composition. If $-dN > 0$ particles evaporate, they carry away a composition of x_A^v representing the vapor in equilibrium. The total change in species A in the liquid is $d(Nx_A) = x_A^v dN$. Expanding $d(Nx_A) = Ndx_A + x_A dN = x_A^v dN$, we rearrange to get:

$$Ndx_A = (x_A^v - x_A)dN \implies dx_A = -(x_A^v - x_A) \left(\frac{-dN}{N} \right)$$

Because of the tie line property derived previously, $T_{\text{dew}}(x_A^v) = T_{\text{bubble}}(x_A)$, meaning $(x_A^v)^2 = x_A(2 - x_A) \implies x_A^v = \sqrt{2x_A - x_A^2}$. Substituting this gives:

$$dx_A = - \left[(2x_A - x_A^2)^{1/2} - x_A \right] \left(\frac{-dN}{N} \right)$$

6. **Integration:** From part (e), $\frac{dN}{N} = \frac{dx_A}{\sqrt{x_A(2-x_A)} - x_A}$. Use substitution $x_A = 1 - \cos \theta$. Then $dx_A = \sin \theta d\theta$ and $\sqrt{x_A(2-x_A)} = \sin \theta$.

$$\frac{dN}{N} = \frac{\sin \theta}{\sin \theta + \cos \theta - 1} d\theta$$

Using half-angle substitution $t = \tan(\theta/2)$, we get $\sin \theta = \frac{2t}{1+t^2}$ and $\cos \theta = \frac{1-t^2}{1+t^2}$, $d\theta = \frac{2dt}{1+t^2}$. The integrand simplifies remarkably:

$$\int \frac{dN}{N} = \int \frac{2dt}{(1-t)(1+t^2)} = \int \left(\frac{1}{1-t} + \frac{1+t}{1+t^2} \right) dt = -\log(1-t) + \frac{1}{2} \log(1+t^2) + \arctan(t)$$

Integrating from N_0 to N , and mapping to x_A : $\log \left(\frac{N}{N_0} \right) = \left[\log \frac{\sqrt{1+t^2}}{1-t} + \arctan t \right]_{t_0}^t$. Note $\arctan t = \theta/2$. Initial condition $x_A = 1/2 \implies \cos \theta_0 = 1/2 \implies \theta_0 = \pi/3 \implies t_0 = 1/\sqrt{3}$. Evaluating at t_0 yields $\log \frac{2}{\sqrt{3}-1} + \frac{\pi}{6}$. The implicit relation is:

$$\log \left(\frac{N}{N_0} \right) = \log \frac{\sqrt{1+t^2}}{1-t} + \arctan t - \log \frac{2}{\sqrt{3}-1} - \frac{\pi}{6}$$

7. Half boiled off: Set $N = N_0/2 \implies \log(N/N_0) = -\log 2 \approx -0.693$. We solve the implicit function $\log \frac{\sqrt{1+t^2}}{1-t} + \arctan t - 1.5286 = -0.693$ numerically. By substituting back to θ , the function is $\frac{\theta}{2} - \log(\cos \frac{\theta}{2} - \sin \frac{\theta}{2}) \approx 0.836$. This yields the root $\theta \approx 0.680$ rad. The composition in the liquid is $x_A = 1 - \cos(0.680) \approx 0.223$. The vapor being produced at this instant is $x_A^v = \sin(0.680) \approx 0.629$. **Comparison:** In the closed system, the remaining liquid at half-boil had $x_A \approx 0.293$. The open beaker liquid is much more depleted of A ($0.223 < 0.293$) because the A-rich vapor continuously escapes and is lost from the system entirely.

8. When does $x_A \rightarrow 0$? Set $x_A \rightarrow 0 \implies \theta \rightarrow 0 \implies t \rightarrow 0$. Substituting $t = 0$ into the implicit relation:

$$\log \left(\frac{N}{N_0} \right) = \log(1) + 0 - 1.5286 = -1.5286 \implies \frac{N}{N_0} = e^{-1.5286} \approx 0.217$$

Since $N/N_0 = 0.217 > 0$, the mole fraction of A reaches strictly zero **before** the boiling is finished. About 21.7% of the pure initial liquid remains when the last trace of A leaves.

9. Why the diff eq fails: The differential equation assumes phase coexistence governed by both the bubble and dew point curves, which implicitly requires the existence of a mixture (both A and B species). Once $x_A = 0$, species A is entirely exhausted. The remaining 21.7% of the liquid is purely species B. At this point, composition no longer changes ($x_A \equiv 0$). The remaining process is simply a standard pure-substance phase transition of pure B at its constant boiling temperature T_0 , which needs to be described simply by latent heat equations, not mixture composition shifts.

Saturated vapor pressure and the Clausius–Clapeyron equation

A condensed phase of substance A , either a liquid or a solid, is in equilibrium with its vapor. The coexistence pressure is called the **saturated vapor pressure** and is denoted as $P_{\text{sat}}(T)$. Let $q(T)$ be the latent heat per particle for the transition from the condensed phase to the vapor. Let $v_A^{\text{vap}}, v_A^{\text{cond}}$ be the volumes per particle of the vapor and the condensed phase.

1. First suppose that the gas phase consists only of vapor A . Assume $v_A^{\text{vap}} \gg v_A^{\text{cond}}$, and approximate the vapor as an ideal gas. Show that the Clapeyron equation simplifies to

$$\frac{d \log P_{\text{sat}}}{dT} = \frac{q(T)}{T^2}. \quad (1)$$

Further assume that q is approximately independent of T , integrate and show

$$P_{\text{sat}}(T) = P_0 \exp\left(-\frac{q}{T}\right),$$

where P_0 is a constant.

2. Now suppose that the gas phase is an ideal gas mixture consisting of vapor A and an inert gas B . Let p_A and p_B be their partial pressures, so that

$$P_{\text{tot}} = p_A + p_B.$$

Assume that p_B is held fixed. The coexistence condition for substance A is

$$\mu_A^{\text{cond}}(T, P_{\text{tot}}) = \mu_A^{\text{gas}}(T, p_A).$$

Show that the same equation (1) holds for the saturated partial pressure of vapor A :

$$\frac{d \log p_A}{dT} = \frac{q}{T^2}.$$

Thus the result of part (a) remains valid, provided P is interpreted as the partial pressure of vapor A , not the total gas pressure.

Solution

1. The standard Clapeyron equation dictates the slope of the phase coexistence curve as a function of temperature:

$$\frac{dP_{\text{sat}}}{dT} = \frac{L}{T\Delta V} = \frac{q(T)}{T(v_A^{\text{vap}} - v_A^{\text{cond}})}$$

where $q(T)$ is the latent heat per particle, and v_A represents volume per particle. Since the vapor is much less dense than the condensed phase, $v_A^{\text{vap}} \gg v_A^{\text{cond}}$, we approximate the volume change strictly as $\Delta V \approx v_A^{\text{vap}}$. Approximating the vapor as an ideal gas, we use the equation of state $P_{\text{sat}} v_A^{\text{vap}} = k_B T \implies v_A^{\text{vap}} = T/P_{\text{sat}}$ (set-

ting $k_B = 1$ following typical thermodynamics conventions used here). Substituting this back into the Clapeyron equation:

$$\frac{dP_{\text{sat}}}{dT} = \frac{q(T)}{T \left(\frac{T}{P_{\text{sat}}} \right)} = \frac{q(T)P_{\text{sat}}}{T^2}$$

Dividing both sides by P_{sat} yields the simplified differential equation:

$$\frac{1}{P_{\text{sat}}} \frac{dP_{\text{sat}}}{dT} = \frac{d \log P_{\text{sat}}}{dT} = \frac{q(T)}{T^2} \quad (1)$$

If we further assume $q(T) \approx q$ (a constant independent of T), we can integrate directly:

$$\int d \log P_{\text{sat}} = \int \frac{q}{T^2} dT \implies \log P_{\text{sat}} = -\frac{q}{T} + C$$

Exponentiating both sides defines $P_0 = e^C$, giving:

$$P_{\text{sat}}(T) = P_0 \exp \left(-\frac{q}{T} \right)$$

2. When the gas phase is a mixture of vapor A and inert gas B, the total pressure is $P_{\text{tot}} = p_A + p_B$. The equilibrium condition sets the chemical potential of A in the condensed phase equal to the chemical potential of A in the gas phase:

$$\mu_A^{\text{cond}}(T, P_{\text{tot}}) = \mu_A^{\text{gas}}(T, p_A)$$

Because the gas is an ideal gas mixture, the chemical potential $\mu_A^{\text{gas}}(T, p_A)$ behaves identically to that of a pure ideal gas of species A evaluated exclusively at its partial pressure p_A . Taking the total derivative with respect to temperature T on both sides along the coexistence curve:

$$\left(\frac{\partial \mu_A^{\text{cond}}}{\partial T} \right)_{P_{\text{tot}}} + \left(\frac{\partial \mu_A^{\text{cond}}}{\partial P_{\text{tot}}} \right)_T \frac{dP_{\text{tot}}}{dT} = \left(\frac{\partial \mu_A^{\text{gas}}}{\partial T} \right)_{p_A} + \left(\frac{\partial \mu_A^{\text{gas}}}{\partial p_A} \right)_T \frac{dp_A}{dT}$$

By the Maxwell relations $d\mu = -s dT + v dP$, the partial derivatives correspond to partial molar entropies and volumes. Since we are given that p_B is fixed, $dP_{\text{tot}} = d(p_A + p_B) = dp_A$. Substituting these into the differential equation gives:

$$-s_A^{\text{cond}} + v_A^{\text{cond}} \frac{dp_A}{dT} = -s_A^{\text{gas}} + v_A^{\text{gas}} \frac{dp_A}{dT}$$

Rearranging to group the dp_A/dT terms together:

$$(v_A^{\text{gas}} - v_A^{\text{cond}}) \frac{dp_A}{dT} = s_A^{\text{gas}} - s_A^{\text{cond}}$$

The change in entropy per particle upon vaporization is identically related to the latent heat via $s_A^{\text{gas}} - s_A^{\text{cond}} = q/T$. Substituting this gives the exact analog to the Clapeyron equation:

$$\frac{dp_A}{dT} = \frac{q}{T(v_A^{\text{gas}} - v_A^{\text{cond}})}$$

Applying the same approximations as before ($v_A^{\text{gas}} \gg v_A^{\text{cond}}$ and $v_A^{\text{gas}} = T/p_A$), we find:

$$\frac{dp_A}{dT} = \frac{q \cdot p_A}{T^2} \implies \frac{d \log p_A}{dT} = \frac{q}{T^2}$$

This mathematically demonstrates that the same Clausius-Clapeyron equation perfectly governs the *partial pressure* of the solvent vapor, independently of the total ambient pressure artificially inflated by the inert gas.